A Methodological Warning for FMAP-Kink Regression Designs: ACA-Expansion Contamination of the Statutory Running Variable in a Public-Data Replication of Leung (2022)

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Structured Abstract

Background. A growing applied public-finance literature uses a regression kink at the 50% statutory Federal Medical Assistance Percentage (FMAP) floor as a regression kink design (RKD) to estimate state Medicaid spending responses to federal generosity.1-3 These applications increasingly span the post-2014 era in which the Affordable Care Act (ACA) Medicaid expansion introduced a 90% enhanced federal match for newly eligible adults in expansion states only — a second margin of federal generosity whose adoption is endogenous to state characteristics that correlate with the FMAP-kink running variable.

Objectives. To document that the pooled FMAP-kink RKD on modern panels is structurally sensitive to ACA-expansion contamination, and to recommend a pre-specified robustness protocol for future applications.

Research Design. Sharp regression kink design at the 50% FMAP floor with state and fiscal-year fixed effects, triangular kernel, MSE-optimal bandwidth, state-clustered standard errors, and enhanced-FMAP episode drops as robustness.

Subjects. Public state-year panel for 50 states plus the District of Columbia, federal fiscal years 2008–2023 (N = 816), built from MACPAC MACStats, HHS Federal Register notices, Bureau of

Economic Analysis (BEA) per-capita personal income, and CMS/Mathematica Medicaid long-term services and supports (LTSS) reports.

Measures. Log total Medicaid spending per capita, log home- and community-based services (HCBS) and institutional LTSS spending per capita, and HCBS share of LTSS dollars.

Results. Dropping ACA-expansion state-years flips the aggregate FMAP-kink response on log total Medicaid per capita from +1.49 (95% CI, −1.06 to 4.03) to −2.53 (95% CI, −4.26 to −0.79) and the HCBS-share response from −0.19 to −1.19 (95% CI, −2.04 to −0.34). An extended-window check (FY2002–FY2023) confirms the flip is structural. The non-expansion subsample is observably unbalanced within bandwidth (2.6× as Southern; 35% vs 13%).

Conclusions. The FMAP-kink RKD on panels spanning the post-2014 expansion era pools two distinct federal-match variations. Future analyses should pre-specify “drop all ACA-expansion state-years” as a primary robustness check and should not treat the non-expansion estimate as cleaner without defending RKD smoothness in the restricted sample.

Key Words: Medicaid; Federal Medical Assistance Percentage; long-term services and supports; regression kink design; ACA Medicaid expansion.

Introduction

The federal-state matching structure of Medicaid creates state-level variation in the marginal price of a Medicaid dollar that an applied public-finance literature has begun to exploit. Three published applications use the regression kink at the 50% statutory Federal Medical Assistance Percentage (FMAP) floor as a quasi-experimental source of variation in federal generosity. Kumar1 uses the kink to estimate the effect of federal Medicaid generosity on household income. Giertz and Kumar2 use it to estimate a local fiscal multiplier of federal-to-state grants. Leung3 applies the kink directly to state Medicaid spending and estimates a FMAP elasticity of 1.6–2.9. The FMAP-kink RKD is increasingly a workhorse design for questions about how state Medicaid spending and downstream outcomes respond to federal financing.

This paper documents a structural problem with that design when it is implemented on panels that span the post-2014 ACA Medicaid expansion era. In a public-data replication of the Leung3 FMAP-kink RKD on federal fiscal years 2008–2023, dropping the ACA-expansion state-years flips the aggregate spending response from +1.49 to −2.53, the home- and community-based services (HCBS) share of long-term services and supports (LTSS) response from −0.19 to −1.19, and the institutional LTSS spending response from +1.83 to +2.34. An extended-window specification check (FY2002–FY2023) holding the ACA drop fixed recovers essentially the same negative non-expansion coefficient on a panel that is 45% larger. The sign flip is structural — it reflects a difference in the underlying state-year response pattern between expansion and non-expansion states, not a sample-size artifact of the post-2014 window. The paper’s primary contribution is to surface that contamination as a methodological observation about the FMAP-kink design and to argue that future analyses spanning the post-2014 era should pre-specify “drop all ACA-expansion state-years” as a primary robustness check rather than a secondary one.

The mechanism is worth stating up front. The ACA Medicaid expansion’s 90% enhanced match for newly eligible adults is *not* a function of the statutory FMAP formula. It is a flat statutory rate applied on top of the statutory FMAP for the expansion population in expansion states only.

Whether a state expanded is a function of state politics and budget structure, and those factors correlate with the running variable. Expansion states are concentrated among non-floor states; non-expansion states are concentrated in the Deep South and among mid-income states below but not at the floor. In the post-2014 panel the expansion indicator is strongly correlated with the running variable, and the expansion state-years responding to the 90% enhanced match are not the same state-years as those near the statutory kink. Pooling the two pulls two distinct federal-match responses through one estimator. Dropping expansion state-years isolates the statutory slope change in a non-expansion subsample whose own selection on observables is, as we show below, structural rather than incidental. The non-expansion estimate is therefore not a "cleaner" preferred estimand. It is a warning signal that the pooled estimate is reaching across a discontinuity in the policy environment that the RKD smoothness conditions are not designed to bridge.

The paper carries two secondary contributions. First, the HCBS-versus-institutional decomposition we originally designed the paper around returns confidence intervals too wide to distinguish small-effect from proportional-expansion explanations on every outcome. We report this null honestly and discuss two competing interpretations; we neither claim to have measured the absence of a federally financed LTSS rebalancing effect nor that we have detected one. Second, the aggregate FMAP-spending response at the kink (FMAP elasticity 0.74; 95% CI −0.53 to 2.02) is directionally consistent with Leung's published 1.6–2.9 range — her lower bound sits inside our upper tail — but too imprecise to adjudicate. Whether her published estimate is itself sensitive to the same ACA-expansion contamination we document is an open empirical question we cannot answer without re-running her specification on her data. A prerequisite contribution is a data pipeline integrity check: because the statutory FMAP is a deterministic function of the running variable, recovering the formula-implied slope change of 0.9487 from the empirical data (we get 0.88–1.04 across bandwidths) is evidence that the MACPAC Exhibit 6, HHS Federal Register, and BEA SAINC1 series are parsed consistently against the §1905(b) statutory formula. This is a necessary data-integrity check, not a behavioral first stage.

Background

The FMAP formula and its 50% floor

The federal government pays a state-specific share of each state's Medicaid benefit expenditures, computed annually by HHS as FMAP = 1 − 0.45 × (state-to-national PCPI ratio)^2, bounded at [0.50, 0.83], with a three-year lag in per-capita personal income so that the FMAP for fiscal year t is computed using calendar years t−4 through t−2.4 The formula is deterministic: given BEA's state-level PCPI vintage, the FMAP for the coming federal fiscal year is mechanical up to the floor and ceiling. Because FMAP is a deterministic function of the running variable, the "first stage" in any FMAP-kink RKD is a formula identity rather than a behavioral relationship — recovering the formula-implied slope change on a real data build is evidence that the data pipeline has been parsed correctly, not evidence that state behavior responds at the kink.

The statutory floor generates a regression kink at *r** √(0.5/0.45) 1.0541. For $r <$ *r** the formula has slope −0.9 · r; for r *r** the floor binds and the slope is zero. At the kink, the mechanical slope change is 0.9 · *r** 0.9487. The 10–14 states at the floor in any given year include California, Connecticut, Maryland, Massachusetts, Minnesota, New Hampshire, New Jersey, New York, Virginia, and Washington, with a rotating set of marginal entrants and exiters (Alaska, Colorado, Illinois, North Dakota, Rhode Island, Wyoming). The running variable is a three-year-lagged ratio of aggregate income statistics that no state government can plausibly manipulate at the margin. Under the Card, Lee, Pei, and Weber5 RKD identification conditions, the ratio of the slope change in the outcome to the slope change in the treatment at the kink identifies a local weighted average of the marginal effect.

Three enhanced-FMAP episodes overlay the statutory schedule

In the modern panel, three federally financed enhanced-match episodes overlay the statutory FMAP schedule. None alters the *location* of the 50% floor in the formula, but each introduces a second margin of federal generosity that potentially contaminates the spending response a researcher would attribute to the statutory FMAP. The American Recovery and Reinvestment Act §5001 bump (FY2009 Q4 – FY2011 Q2) provided a temporary across-the-board FMAP increase of 6.2 percentage points plus an unemployment-keyed top-up. The Affordable Care Act Medicaid expansion (effective in expansion states from January 2014 onward) provides a 90% enhanced federal match for newly eligible adults — distinct from, and applied on top of, the statutory FMAP for the traditional Medicaid population. The Families First Coronavirus Response Act §6008 bump (FY2020 Q1 through FY2023 Q1, with phase-down through December 2023) provided a 6.2-percentage-point across-the-board FMAP increase conditional on continuous enrollment.6,7 The ARRA and FFCRA episodes are temporary and apply uniformly to all states; the ACA expansion is permanent, applies only in expansion states, and is the methodologically consequential episode for the FMAP-kink design.

Medicaid LTSS rebalancing

Medicaid is the single largest payer for long-term services and supports in the United States, financing roughly 42% of national LTSS spending.8 Through the 1980s institutional care dominated Medicaid LTSS spending; a series of federal policy actions — the 1915(c) waiver authority, the *Olmstead* decision, Money Follows the Person, the Balancing Incentive Program, and the ACA §2401 1915(k) Community First Choice option — expanded state authority to deliver LTSS in home and community settings. By FY2022 the Medicaid and CHIP Payment and Access Commission (MACPAC) reports 65% of Medicaid LTSS dollars flow to HCBS.9,10 Whether HCBS expansion *substitutes* for institutional care has been the central empirical question, and the literature's answer is "partially." Guo, Konetzka, and Manning11 identify a causal substitution effect via claims-based instrumental variables; McGarry and Grabowski12 estimate that each dollar of HCBS spending is offset by approximately $0.26 of institutional Medicaid LTSS spending. Konetzka13 cautions that HCBS cost-effectiveness is not robust to severity progression, cognitive impairment, and unmet need. Lieber and Lockwood14 document that in-kind home care is less valued than its cash-equivalent cost but that welfare gains from targeting dominate. Unmet need persists in the form of Medicaid HCBS waiver waiting lists: as of 2024, forty states maintained waiting or interest lists for HCBS waivers and over 710,000 people were on them, three-quarters of them individuals with intellectual or developmental disabilities.15 None of this literature identifies rebalancing from the federal-financing margin. The conceptual gap we attempted to fill was whether the mechanical variation in federal match at the FMAP floor translates into a measurable shift in the HCBS-institutional composition of state Medicaid LTSS spending. As we show in the Results section, that question is not identified at the precision available on public state-year data. The HCBS decomposition is therefore reported as secondary evidence that the aggregate FMAP-kink response does not decompose cleanly within LTSS.

Methods

Panel and sources

The analytic file is a state-year panel indexed by state and federal fiscal year, covering the 50 states and the District of Columbia for FY2008–FY2023 ($N = 816$). Statutory FMAP values come from the MACPAC MACStats Exhibit 6 archive (editions March 2011 through December 2024), cross-validated against HHS Federal Register notices for eight fiscal years from govinfo.gov; across 357 overlapping state-years mean $|\text{FR} - \text{MACStats}| = 0.003$ percentage points, and the only systematic disagreement is Louisiana’s Katrina/Rita adjustments.[4,16] State personal income comes from BEA SAINC1 spanning calendar years 1929–2024. For each fiscal year t we compute state PCPI three-year average over calendar years t−4 through t−2 divided by the national analog, matching the HHS lag structure. The centered running variable is $\tilde{r}$s,t = rs,t − *r**.

HCBS and institutional LTSS spending totals come from the CMS/Mathematica Medicaid LTSS Annual Expenditures Reports.[10] We harmonize spending categories into an HCBS bucket (1915(c), (i), (j), (k); state plan personal care; state plan home health; PACE; MFP; and 1115 HCBS lines) and an institutional bucket (nursing facility, ICF/IID, mental-health-facility LTSS). Post-2017 Mathematica vintages back-cast a harmonized HCBS allocation that conflicts with contemporaneous pre-2017 vintages; we use the latest harmonized vintage throughout. The FY2015 and FY2016 Mathematica vintages are PDF-only and are dropped from the LTSS outcome panel (leaving 713 LTSS state-years of the 816 FMAP panel). Extending the panel back to FY2002 collapses the HCBS spending point estimate from 1.18 to 0.01, consistent with either time-heterogeneity in the response or with pre-2017 vintage harmonization drift — we treat the FY2008–FY2023 HCBS point estimate as conditional on the post-2017 harmonized vintage being correctly back-cast, and we flag this as a standing limitation in the Discussion. HCBS-user-share outcomes from CMS T-MSIS Analytic Files briefs for FY2019–FY2023 are reported as an appendix cross-check rather than a co-equal main outcome.

Empirical strategy

We estimate a sharp regression kink design at the 50% statutory FMAP floor.[5,17] Within a bandwidth h around the kink, we estimate the local-quadratic regressions

Ys,t = Y + Y− · $\tilde{r}$s,t · [$\tilde{r}$s,t < 0] + Y+ · $\tilde{r}$s,t · [$\tilde{r}$s,t 0] + Y · $\tilde{r}$s,t2 + Y Xs,t + s,tY

(and the analog for FMAP) on a triangular-kernel-weighted sample, with state and fiscal-year fixed effects in Xs,t and the sharp RKD estimand RKD = (Y+ − Y−) / (F+ − F−). The bandwidth is MSE-optimal following Calonico, Cattaneo, and Titiunik.[17] Standard errors are clustered by state. We use local-quadratic ($p = 2$) as the baseline polynomial order; CCT recommend $p = p +$ 1 for bias correction, with the point estimate from $p = 2$ and the robust confidence interval from a $p = 3$ fit on the same bandwidth. Pure $p = 3$ at half the MSE-optimal bandwidth produces an obvious outlier on institutional LTSS (= −14.7) from an underidentified local cubic on a small sample, consistent with CCT’s warning about cubic instability at narrow bandwidths in sparse panels.

Our reduced-form coefficient on a log-outcome regression is a semi-elasticity with respect to FMAP expressed as a fraction on [0, 1]. To compare with Leung’s[3] elasticity convention we multiply the semi-elasticity by the FMAP value at the kink (0.50); the translation turns a semi-elasticity of 1.49 on log(total Medicaid/pop) into a FMAP elasticity of 0.74. The running variable uses statutory FMAP and each enhanced-FMAP episode is handled as a robustness drop, not a baseline modification.

Results

Data pipeline integrity check (formula reconstruction of the FMAP first stage)

Table 1 reports the empirical slope change in statutory FMAP at the 50% floor. Across bandwidth choices, the empirical slope change is 0.88–1.04, bracketing the mechanical 0.9487 implied by differentiating the formula at *r **= 1.054. The small residual deviation traces to D.C.'s 70% statutory override and to Louisiana Katrina/Rita adjustments.** Because the statutory FMAP is a deterministic function of the running variable, this recovery speaks to data pipeline integrity — it confirms that MACPAC Exhibit 6, the HHS Federal Register notices, and BEA SAINC1 are being parsed consistently against the §1905(b) statutory formula — and is a necessary prerequisite rather than a behavioral validation.** We present this result first as a data-integrity check, not as the paper's headline, and we explicitly do not characterize it as a "first stage replication."

Table 1. FMAP first-stage formula reconstruction. Empirical slope change in statutory FMAP at the 50% floor, by bandwidth.

Aggregate FMAP-kink response on public data

On log total Medicaid per capita, the baseline RKD point estimate at MSE-optimal bandwidth is = 1.488 (SE 1.299), 95% CI −1.06 to 4.03 on 652 state-years and 45 state clusters. Translating to Leung's3 elasticity convention via multiplication by FMAP = 0.50 gives a FMAP elasticity of 0.74 (95% CI −0.53 to 2.02). Leung's published 1.6–2.9 range sits in the upper half of our confidence interval: her lower bound (1.6) is below our upper bound (2.02), but her point estimate is above ours by roughly two to four, and our interval crosses zero. We characterize this as directionally consistent in the limited sense that her published lower bound (1.6) is below our upper bound (2.02); our point estimate is approximately half her lower bound, and our 95% interval crosses zero. It is too imprecise to adjudicate Leung's3 headline.

HCBS-versus-institutional decomposition (secondary evidence)

Table 2. HCBS-versus-institutional decomposition at MSE-optimal bandwidth.

Outcome		SE	95% CI	N
log(HCBS $ / pop)	1.183	1.273	−1.31 to 3.68	623
log(institutional $ / pop)	1.831	1.891	−1.88 to 5.54	623
HCBS share of LTSS $	−0.194	0.556	−1.28 to 0.90	623

None of the three outcomes is distinguishable from zero at the 5% level. We report point estimates and confidence intervals only and do not interpret the directional pattern as a finding — the standard errors are far too wide. The HCBS user share from CMS TAF (FY2019–FY2023) is reported in the appendix as a short-window cross-check and is not treated as a co-equal main outcome. The interpretation of why the decomposition does not show up — and what it would take to recover an effect — is in the Discussion.

ACA-expansion contamination: the headline finding

The paper's headline is that the FMAP-kink RKD on this public-data panel is structurally not robust to dropping ACA-expansion state-years. Table 3 reports the enhanced-FMAP episode drops; the ACA-expansion row is the load-bearing one.

Table 3. Enhanced-FMAP episode drops, by outcome. 95% CIs in brackets.

Specification	log(total Medicaid / pop)	log(HCBS / pop)	log(institutional / pop)	HCBS share	N
Baseline (FY08–FY23)	1.488 [−1.06, 4.03]	1.183 [−1.31, 3.68]	1.831 [−1.88, 5.54]	−0.194 [−1.28, 0.90]	652
Drop ARRA (FY09–FY11)	0.827 [−1.51, 3.16]	0.594 [−1.90, 3.08]	1.666 [−2.83, 6.16]	−0.321 [−1.48, 0.84]	576
Drop ACA expansion	**−2.526** [−4.26, −0.79]	**−3.018** [−5.36, −0.68]	**2.336** [0.20, 4.48]	**−1.191** [−2.04, −0.34]	417
Drop FFCRA (FY20–FY23)	1.259 [0.04, 2.48]	0.371 [−1.77, 2.51]	1.821 [−2.16, 5.80]	−0.303 [−1.31, 0.70]	481
Drop all three	−2.306 [−4.21, −0.40]	−2.946 [−5.73, −0.16]	2.932 [−1.23, 7.09]	−1.356 [−2.71, −0.001]	240

The ARRA drop and the FFCRA drop move the aggregate by at most one standard error — ordinary episode-sensitivity perturbations. The ACA-expansion drop is different. Dropping every state-year in which the state is operating under the ACA Medicaid expansion's 90% enhanced match moves the aggregate from +1.49 to −2.53, with a 95% confidence interval that now excludes zero. The HCBS spending coefficient moves from +1.18 to −3.02, institutional spending from +1.83 to +2.34 (lower bound now excludes zero), and the HCBS share of LTSS dollars from −0.19 to −1.19. Dropping all three episodes jointly reinforces this pattern. In the non-expansion subsample, a higher federal match rate is associated with lower total Medicaid spending per capita, higher institutional LTSS spending per capita, and a lower HCBS share of LTSS — in the opposite direction from the prospective rebalancing mechanism.

Is the sign flip a small-sample artifact? To rule out a small-sample driver, we extended the panel back to FY2002 (adding 306 state-years) and re-ran the ACA-expansion drop. The extended-window log(total Medicaid / pop) non-expansion is −2.60 [−4.26, −0.95] on 606 state-years — essentially identical to the baseline-window −2.53 on 417 state-years. The sign flip is **structural**, not sample-size-driven. (The HCBS spending point estimate in the same extended-window check moves from 1.18 to 0.01; we treat that vintage-sensitivity finding separately in the Discussion.)

Is the non-expansion subsample observably balanced? It is not. Table 4 reports observable means for expansion and non-expansion state-years within the MSE-optimal bandwidth of the running variable (|r − *r**| 0.230). State-year-level means; no standard errors, no differences, no inference.

Table 4. Observable balance, expansion vs non-expansion state-years near the kink. FY2008–FY2023; N = 652 state-years within bandwidth (275 expansion, 377 non-expansion). Computed by `analysis/balance_check_mr1.py`.

Variable	Expansion	Non-expansion
State population (millions)	6.82	7.21
Share age 65+	0.163	0.141
Pre-2014 mean HCBS share of LTSS	0.494	0.484
Share South (Census region)	0.135	0.350
Share Northeast	0.247	0.138
Share Midwest	0.280	0.305
Share West	0.338	0.207
Running variable (state/US PCPI ratio, 3-yr lag)	1.011	0.980
Centered running variable (r − *r**)	−0.043	−0.074
Share of state-years at the FMAP floor	0.364	0.233

Non-expansion state-years near the kink are concentrated more than 2.6 times as heavily in the South as the expansion state-years (35.0% vs 13.5%), under-represent the Northeast and West, and have a slightly older population. Floor-binding states within the bandwidth are roughly the same set in both subsamples (14 states each, with substantial overlap including CA, NY, NJ, MA, MD, MN, NH, VA, WA, IL, CO, AK); the subsamples differ in *which years* of each state's panel contribute to the non-expansion flag. Non-expansion state-years are dominated by the pre-2014 era for states that subsequently expanded plus the entire post-2014 era for states that never expanded; expansion state-years are post-2014 only. **We acknowledge that non-expansion status is itself endogenous and correlated with state characteristics that may independently affect the running variable and the outcome. The non-expansion should not be read as a causal estimate of the FMAP response in the counterfactual world where no state expanded.** It should be read as a warning signal that the pooled estimator is reaching across a discontinuity in the policy environment that the RKD smoothness conditions are not designed to bridge.

Sensitivity analyses

The remaining checks are reported in full in the appendix. *Bandwidth sensitivity.* At 0.5×, 1.0×, and 2.0× the MSE-optimal bandwidth and at both $p = 2$ and $p = 3$, every 95% CI on the decomposition outcomes contains zero. At double the MSE-optimal bandwidth, the total-Medicaid attenuates to 0.70 [−0.26, 1.65] — a tighter CI but a smaller point estimate, which cuts in the opposite direction from a "precision is structural" narrative. *Leave-one-state-out.* For log total Medicaid per capita the range across 51 re-estimates is 2.13 against a baseline of 1.49 — the classic signature of a small-effective-sample-near-the-floor design, with the median leave-one-out estimate close to the baseline for every outcome. *Placebo kinks.* Of 20 non-true-kink placebo estimates at ±0.02 and ±0.05, only one 95% CI excludes zero, consistent with expected false-positive rates. *HCBS-share robustness menu.* Drop-D.C., local cubic, drop-CFC, pop-65+ denominator, and donut specifications all return point estimates that are not statistically distinguishable from the baseline −0.19.

Discussion

What the ACA sign flip does and does not imply

The ACA sign flip is a methodological observation about pooling, not a substantive identification of a "non-contaminated" FMAP response.

What the sign flip does imply. The pooled FMAP-kink estimator on the FY2008–FY2023 panel is sensitive to whether expansion state-years are included. The sensitivity is not a small-sample artifact (the extended-window check rules that out) and it is not an ordinary episode-sensitivity perturbation of the kind ARRA and FFCRA produce. It is large enough to flip the sign of three of four primary outcomes and to push two of them outside zero on the side opposite the prospective rebalancing mechanism. Any FMAP-kink RKD spanning the post-2014 era should expect this kind of sensitivity and should pre-specify the expansion drop as a primary robustness check.

What the sign flip does not imply. The non-expansion subsample is not a counterfactual world without expansion. It is a non-randomly selected subset of state-years whose observable characteristics (Table 4) differ from the full panel in the South-vs-non-South direction, in regional composition, and in age structure. Non-expansion status is itself endogenous to state political economy and budget structure. We do not have a separate identification argument for why the smoothness conditions of the RKD should hold within the non-expansion sample, and we do not offer one. Two specific concerns: (i) Southern non-expansion states have systematically higher institutional LTSS shares than Northeastern and West-Coast states,9,10 so a regional composition shift in the non-expansion subsample could mechanically produce the observed institutional-LTSS sign without any FMAP-driven response; and (ii) the non-expansion subsample is disproportionately a pre-ACA-era panel for the states near the kink, so the response we measure is a mix of a pre-2014 statutory FMAP slope and a post-2014 slope from the never-expanding states only — two policy-era subsamples that may have different underlying response functions.

The honest summary is narrow: the FMAP-kink estimator on this panel is not robust to expansion-episode inclusion, and the non-expansion subsample is observably selected. Both observations are methodologically consequential. Neither identifies a "true" non-contaminated FMAP response.

Implications for FMAP-kink research design

The methodological recommendation for future FMAP-kink research is concrete:

> **Recommendation.** Analyses using the FMAP-kink RKD over panels that include the post-2014 ACA-expansion era should pre-specify "drop all ACA-expansion state-years" as a **primary** robustness check, with baseline and non-expansion estimates reported side by side. If the two move by less than one standard error of the baseline, the analyst can interpret the baseline as a pooled estimate. If the two move by more than one standard error (as in our Table 3), the analyst owes the reader an explicit discussion of which subsample is the right estimand for the question at hand. **When the two estimates differ materially, the non-expansion subsample should not be treated as a "cleaner" estimand without a separate demonstration that the smoothness conditions for the RKD hold in the restricted sample** — including a balance check on observable state characteristics within the bandwidth, and an explicit acknowledgment of which states' state-years are carrying the non-expansion estimate.

This recommendation does not require a researcher to *solve* the ACA-contamination problem — it requires only that the researcher transparently document the contamination and bound how much of the headline estimate is attributable to it.

Why the HCBS decomposition does not show up

Two explanations are consistent with the null decomposition, and our data cannot distinguish them. First, the true effect may be small enough that an RKD on public state-year panel data cannot detect it. The McGarry-Grabowski12 \$0.26 institutional offset per HCBS dollar implies a reduced-form response on HCBS share at a small marginal federal subsidy on the order of 1 percentage point or less; our baseline HCBS-share SE of 0.56 is an order of magnitude too large to detect such effects. Second, the FMAP response may be truly concentrated in aggregate Medicaid spending without a distinct LTSS composition effect. A proportional expansion of all Medicaid service categories would be consistent with both Leung's3 aggregate elasticity and our decomposition null. The two explanations are not mutually exclusive; a world in which the true effect is small and the response is partly proportional within LTSS is also consistent with the null. We are not claiming the absence of a rebalancing effect — we are claiming the data do not identify one, and that the counterfactual worlds under each explanation imply different research designs to recover an effect.

Comparison to Leung (2022)

Leung3 estimates a FMAP elasticity of 1.6–2.9 on per-beneficiary state Medicaid spending. Our aggregate FMAP elasticity at the kink, 0.74 [−0.53, 2.02], is directionally consistent in the limited sense that her published lower bound (1.6) is below our upper bound (2.02), but our point estimate is approximately half hers and our interval contains zero. Two observations on the precision gap. First, Leung uses a longer panel that predates the modern enhanced-FMAP episodes, which may yield tighter standard errors. Second, whether our methodological warning applies to Leung's own identifying variation is an open empirical question that we cannot answer without re-running her specification with a pre-specified ACA-expansion drop on her data. We flag this as a useful direction for future replication work and do not attempt the comparison from our public-data build alone.

Limitations

Six limitations structure the interpretation. *First*, the effective sample near the 50% floor is modest (620–650 state-years and 45–49 state clusters at MSE-optimal bandwidth, with only 10–14 unique floor-binding states). Precision is not solvable without a longer panel or a microdata supplement. *Second*, **non-expansion status is endogenous**; the non-expansion subsample is observably selected on regional and political-economy characteristics. *Third*, **vintage-harmonization sensitivity in the HCBS outcome**: extending the panel back to FY2002 collapses the HCBS spending point estimate from 1.18 to 0.01, consistent with either time-heterogeneity in the response or pre-2017 Mathematica vintage harmonization drift. We treat the FY2008–FY2023 HCBS point estimate as conditional on the post-2017 harmonized vintage being correctly back-cast. A reader who builds an independent HCBS panel from contemporaneous pre-2017 vintages may recover a different decomposition coefficient. *Fourth*, CMS-64 and Mathematica LTSS classification changes introduce vintage-discontinuity risk (FY2015 and FY2016 Mathematica vintages are PDF-only and were dropped, leaving 713 LTSS state-years of 816 in the FMAP panel). *Fifth*, LTSS spending outcomes are nominal dollars. *Sixth*, the TAF HCBS user share is an appendix cross-check only and is not a co-equal main outcome. The state-level unit of observation is inherently different from the beneficiary-level LTSS substitution literature and does not translate into welfare conclusions about HCBS as a service delivery model.

Conclusions

On fully public Medicaid data for FY2008–FY2023, the FMAP-kink RKD is structurally not robust to dropping ACA-expansion state-years. Dropping the expansion episode flips the aggregate state Medicaid spending response from +1.49 to −2.53, the HCBS share response from −0.19 to −1.19, and the institutional LTSS response from +1.83 to +2.34, with an extended-window specification check confirming that the flip is structural rather than sample-size-driven. The non-expansion subsample is itself observably selected on regional and political-economy characteristics within the MSE-optimal bandwidth (Table 4) — concentrated 2.6 times as heavily in the South as the expansion subsample — so we do not interpret the −2.53 as a cleaner estimand. We interpret it as a methodological warning that the pooled FMAP-kink estimator on any panel spanning the post-2014 era is reaching across a policy-environment discontinuity the RKD smoothness conditions are not designed to bridge, and we recommend that future FMAP-kink analyses pre-specify "drop all ACA-expansion state-years" as a primary robustness check.

The paper's two secondary contributions are an honest null on the HCBS-versus-institutional decomposition the design was originally meant to test (every confidence interval contains zero, and the data cannot distinguish small-effect from proportional-expansion explanations) and a directionally consistent but imprecise public-data replication of Leung's3 aggregate FMAP elasticity. A prerequisite contribution is the data-pipeline integrity check: recovery of the formula-implied FMAP slope change of 0.9487 (0.88–1.04 empirically) confirms the public-data build against the §1905(b) statutory formula. We are explicit that this last result is a data-integrity check, not a behavioral first-stage replication, because the statutory FMAP is a deterministic function of the running variable and the kink is mechanical by construction.

Three follow-up research directions are implied. First, a regional analysis that explicitly conditions the RKD on Southern versus non-Southern floor states would directly address the load-bearing limitation of the non-expansion subsample — this is the most direct response to our methodological warning. Second, replicating Leung's3 specification with a pre-specified ACA-expansion drop on her own panel would directly test whether her headline 1.6–2.9 elasticity is sensitive to the same contamination we document. Third, a restricted-use T-MSIS Analytic Files supplement with user-level LTSS utilization data could close the precision gap between what public state-year data support and the magnitudes the McGarry-Grabowski12 substitution literature suggests are plausible.

The paper does not claim a rebalancing effect or its absence; it claims that the pooled FMAP-kink design is structurally fragile to expansion-episode inclusion and that future work using the design must confront this contamination explicitly. That is a modest but, we believe, useful contribution to a literature that has begun to lean on the FMAP kink without yet acknowledging what the post-2014 enhanced-match overlay does to the design's identifying variation.

References

1. Kumar A. Does Medicaid generosity affect household income? *Econ Model.* 2020;92:239–256. doi:10.1016/j.econmod.2020.01.004.
2. Giertz SH, Kumar A. The local fiscal multiplier of intergovernmental grants: evidence from federal Medicaid assistance to states. *Int Tax Public Finance.* 2024;31(5):1204–1228. doi:10.1007/s10797-023-09792-y.

3. Leung P. State responses to federal matching grants: the case of Medicaid. *J Public Econ.* 2022;216:104746. doi:10.1016/j.jpubeco.2022.104746.
4. Mitchell A. *Medicaid's Federal Medical Assistance Percentage (FMAP).* Washington, DC: Congressional Research Service; April 2, 2025. CRS Report R43847. https://www.congress.gov/crs-product/R43847.
5. Card D, Lee DS, Pei Z, Weber A. Inference on causal effects in a generalized regression kink design. *Econometrica.* 2015;83(6):2453–2483. doi:10.3982/ECTA11224.
6. Clemens J, Ippolito B, Veuger S. Medicaid and fiscal federalism during the COVID-19 pandemic. *Public Budg Finance.* 2021;41(4):94–109. doi:10.1111/pbaf.12287.
7. Dague L, Ukert B. Pandemic-era changes to Medicaid enrollment and funding: implications for future policy and research. *J Policy Anal Manage.* 2024;43(4):1229–1259. doi:10.1002/pam.22539.
8. Reaves EL, Musumeci MB. *Medicaid and Long-Term Services and Supports: A Primer.* Washington, DC: Kaiser Family Foundation; December 2015. https://www.kff.org/medicaid/medicaid-and-long-term-services-and-supports-a-primer/.
9. Medicaid and CHIP Payment and Access Commission (MACPAC). *Access to Home- and Community-Based Services.* Chapter 4, June 2023 Report to Congress on Medicaid and CHIP. Washington, DC: MACPAC; 2023. https://www.macpac.gov/publication/access-to-home-and-community-based-services-2/.
10. Murray C, Tourtellotte A, Lipson D, Wysocki A. *Medicaid Long-Term Services and Supports Annual Expenditures Report: Federal Fiscal Years 2017 and 2018.* Washington, DC: Mathematica, for the Centers for Medicare & Medicaid Services; January 2021. https://www.medicaid.gov/medicaid/long-term-services-supports/downloads/ltssexpenditures-2017-2018.pdf.
11. Guo J, Konetzka RT, Manning WG. The causal effects of home care use on institutional long-term care utilization and expenditures. *Health Econ.* 2015;24(Suppl 1):4–17. doi:10.1002/hec.3155.
12. McGarry BE, Grabowski DC. Medicaid home and community-based services spending for older adults: is there a "woodwork" effect? *J Am Geriatr Soc.* 2023;71(10):3143–3151. doi:10.1111/jgs.18478.
13. Konetzka RT. The hidden costs of rebalancing long-term care. *Health Serv Res.* 2014;49(3):771–777. doi:10.1111/1475-6773.12190.
14. Lieber EMJ, Lockwood LM. Targeting with in-kind transfers: evidence from Medicaid home care. *Am Econ Rev.* 2019;109(4):1461–1485. doi:10.1257/aer.20180325.
15. Burns A, Wolk A, O'Malley Watts M, Mohamed M, Peña MT. *A Look at Waiting Lists for Medicaid Home- and Community-Based Services from 2016 to 2024.* Washington, DC: Kaiser Family Foundation; October 31, 2024. https://www.kff.org/medicaid/a-look-at-waiting-lists-for-medicaid-home-and-community-based-services-from-2016-to-2024/.
16. Medicaid and CHIP Payment and Access Commission (MACPAC). *Exhibit 6: Federal Medical Assistance Percentages (FMAPs) and Enhanced FMAPs (E-FMAPs) by State, FYs 2022–2025.* MACStats Data Book. Washington, DC: MACPAC; 2024. https://www.macpac.gov/wp-content/uploads/2024/09/EXHIBIT-6-FMAP-and-Enhanced-FMAP-by-State-FYs-2022%E2%80%932025.pdf.
17. Calonico S, Cattaneo MD, Titiunik R. Robust nonparametric confidence intervals for regression-discontinuity designs. *Econometrica.* 2014;82(6):2295–2326. doi:10.3982/ECTA11757.
18. Benchimol EI, Smeeth L, Guttmann A, et al. The REporting of studies Conducted using Observational Routinely-collected health Data (RECORD) Statement. *PLoS Med.* 2015;12(10):e1001885. doi:10.1371/journal.pmed.1001885.

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